

CS & IT ENGINEERING



Computer Network - 2

Transport & Application Layer

Lecture No 05



By - Abhishek Sir



Recap of Previous Lecture



Topic

TCP Timers

Topic

Application Layer

Topic

DNS



Topics to be Covered



Topic

DNS

Topic

HTTP



[MSQ] [GATE-2021, 2-Marks]



#Q. A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?

- ☒ A If the client was waiting to receive a packet, it may wait indefinitely [TRUE]
- ☒ B The TCP server application on S can listen on P after reboot default [False] if startup prog. list entry [TRUE]
- ☒ C If the client sends a packet after the server reboot, it will receive a RST segment if B is true then C is definitely TRUE
- ☐ D If the client sends a packet after the server reboot, it will receive a FIN segment

Ans: A

IITB $\Rightarrow$ MSQ $\Rightarrow$ Ans: A, B, C

TCP client
TCP connection
Socket
[No keepalive timer]

CS-24
IISC
MSQ
only one
option
correct

MSQ $\Rightarrow$ 2, 3
 $\Rightarrow$ ① / $\boxed{4}$

segment

RST

Host S 

TCP server
[Port No. = P]
 $\Rightarrow$ LISTEN state

TCP connection
Socket

Crashed
Reboot
[TCP connection
state lost]

After Reboot
if entry exist in
startup prog. list
 $\Rightarrow$ LISTEN, Port P
No conn. state exists



Topic : DNS



- Domain Name System (DNS)
- 'Stateless' Application Layer Protocol
- Client - Server Model



Topic : DNS



- Used to translate 'Host Name' into 'IP Address' (vice versa)
- Hosts (DNS clients) and DNS server communicate to resolve names
[Name / Address translation]



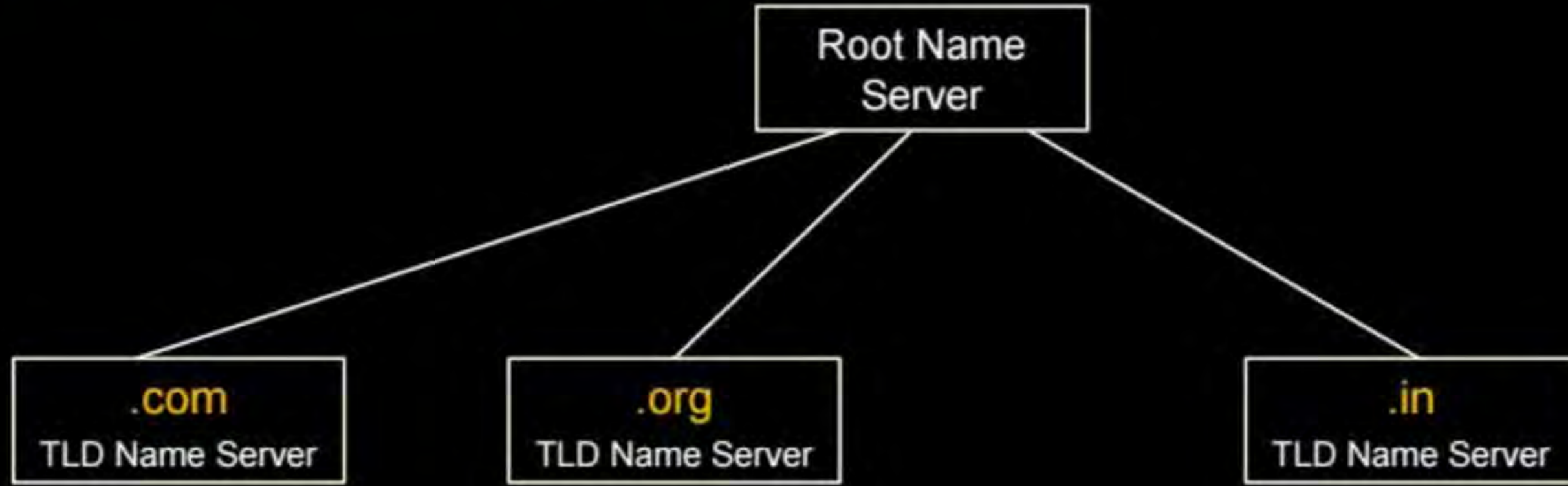
Topic : DNS Hierarchy



- Distributed Hierarchical Database : Global Hierarchy
[Implemented in hierarchy of many name servers]



Topic : DNS Hierarchy





Topic : DNS Hierarchy



→ DNS name servers :

Maintained by DNS service provider

1. Root Name Servers

2. Top-Level Domain (TLD) Name Servers

3. Authoritative Name Servers ✓



Topic : Root Name Servers



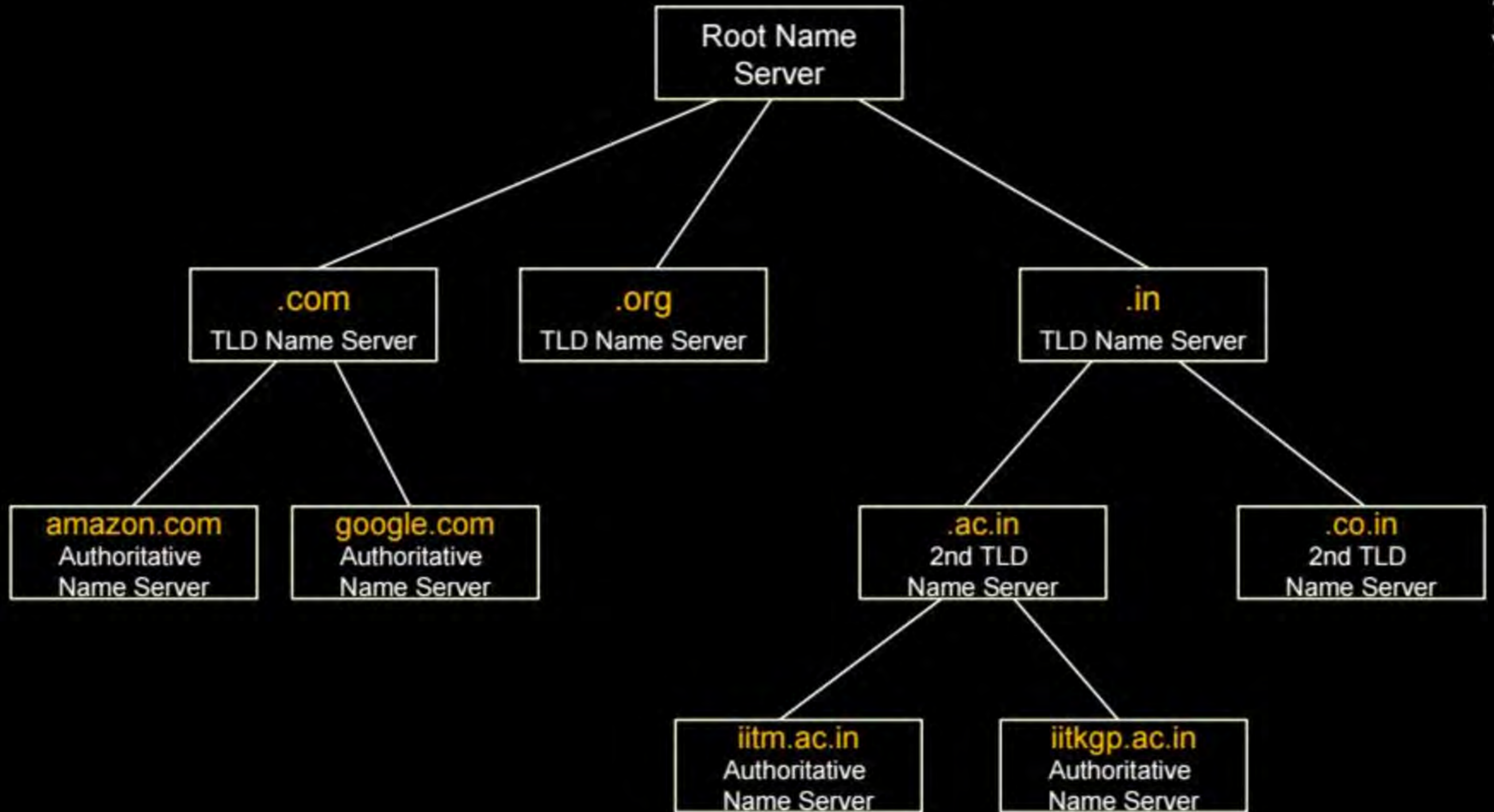
- Point of contact for any DNS query
- ICANN manages root DNS domain
- 13 logical root name servers world wide
[each server replicated many times]



Topic : TLD Name Servers



→ Responsible for all **top level domains** like “.com”, “.org” etc
[**Country domains** e.g. “.in”, “.uk”, etc]





Topic : Authoritative Name Servers

- Organization's own DNS Server(s) ✓
- Providing authoritative host name to IP mappings ✓
- Can be maintained by organization or service provider



Topic : DNS Name Resolution

⇒ Address Translation 

→ DNS Name Resolution or Lookup : 'DNS Resolver Program'

→ Two mode :-

1. Iterated Query
2. Recursive Query



Topic : Iterated Query



→ Contacted server replies with name (address) of server to contact

Example :-

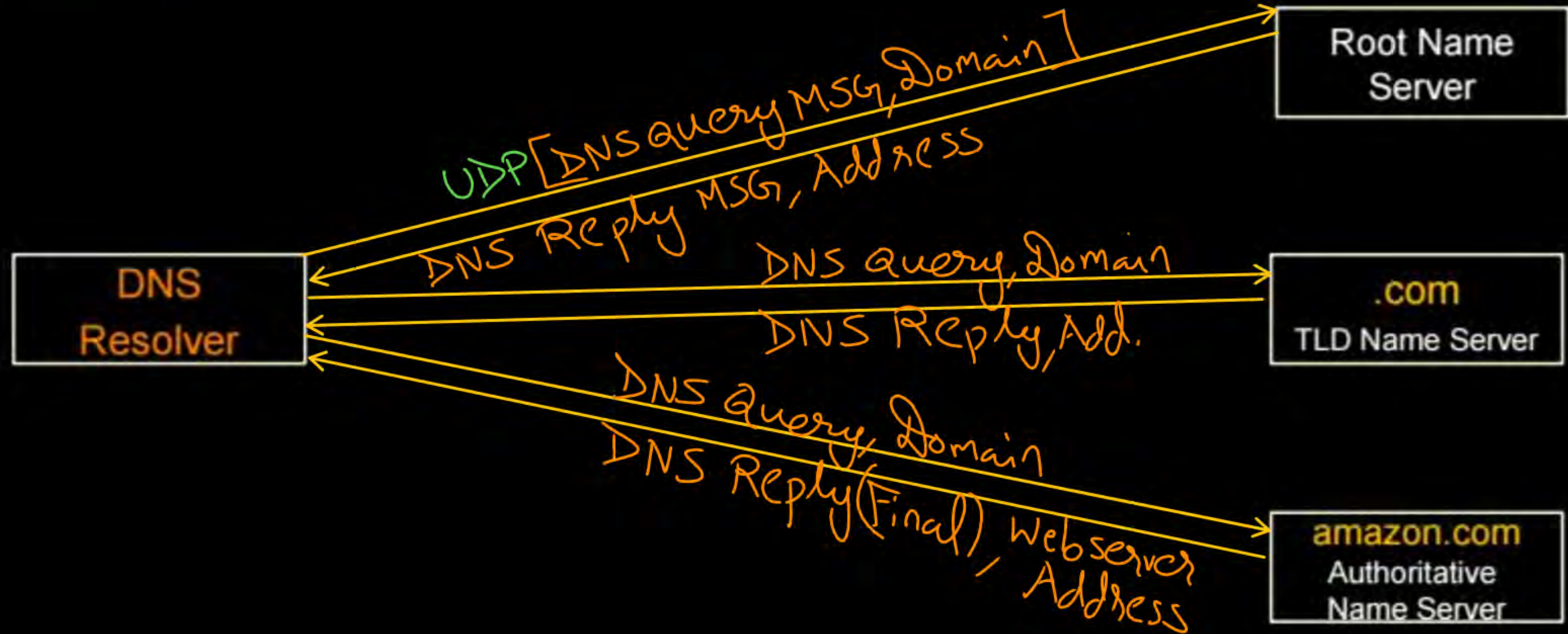
Domain Name : "www.amazon.com"

Domain Name : "www.amazon.com"

Query Mode : Iterative

(3-tier)

DNS Hierarchy



Example 1:-

#Q. Consider the resolution of the domain name "www.amazon.com" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.

Ans = 3

Example 2:-



#Q. Consider the resolution of the domain name 'www.amazon.com' by a DNS resolver. Assume that iterative query mechanism is used in the resolution. Let RTT denote the round trip time between DNS resolver and DNS servers in the DNS hierarchy. The number of RTTs required in completely resolving the domain name is _____.

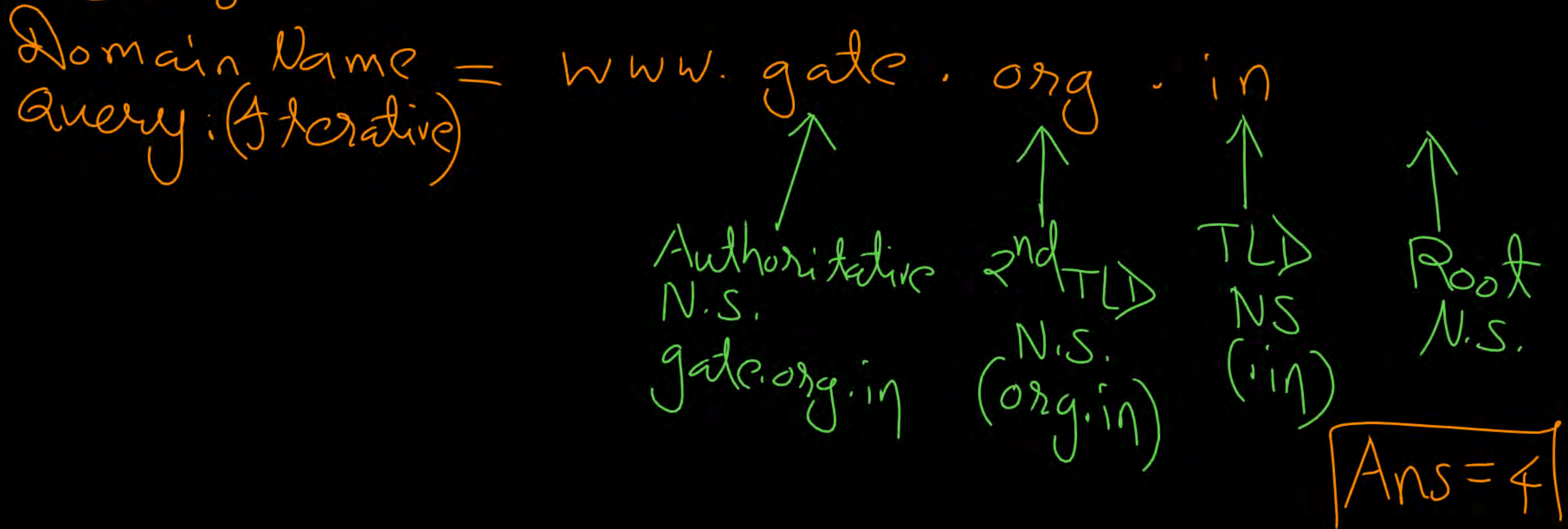
$$\begin{aligned} \text{Ans} &= \text{RTT}_1 + \text{RTT}_2 + \text{RTT}_3 \\ &= 3 \text{RTT} \end{aligned}$$

$$\boxed{\text{Ans} = 3}$$

[NAT] [GATE-2022, 1-Mark] IIT-KGP

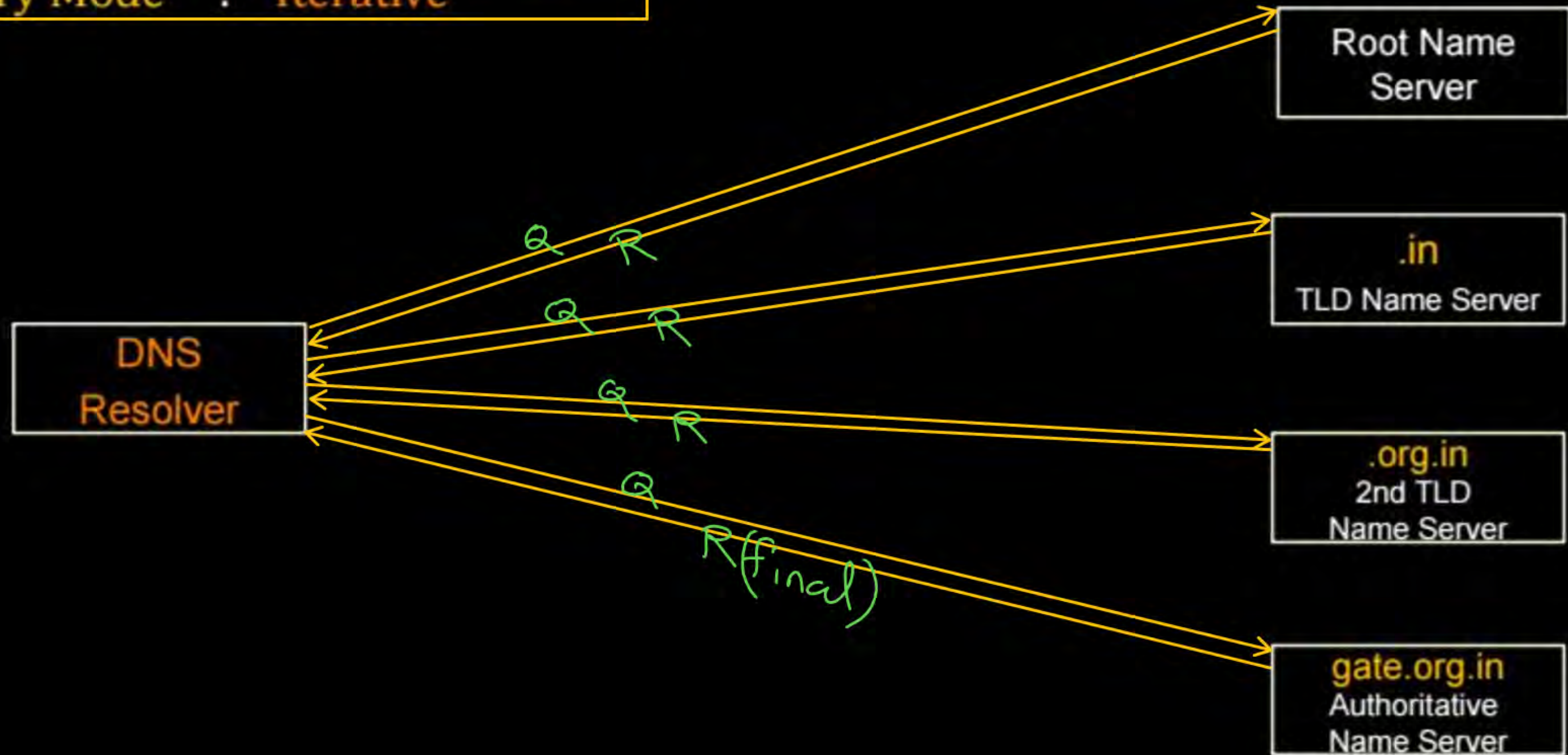


#Q. Consider the resolution of the domain name "www.gate.org.in" by a DNS resolver. [Assume that no resource records are cached anywhere across the DNS servers] and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.



Domain Name : "www.gate.org.in"
Query Mode : Iterative

(4-tier)
DNS Hierarchy





Topic : Recursive Query

→ highly used;



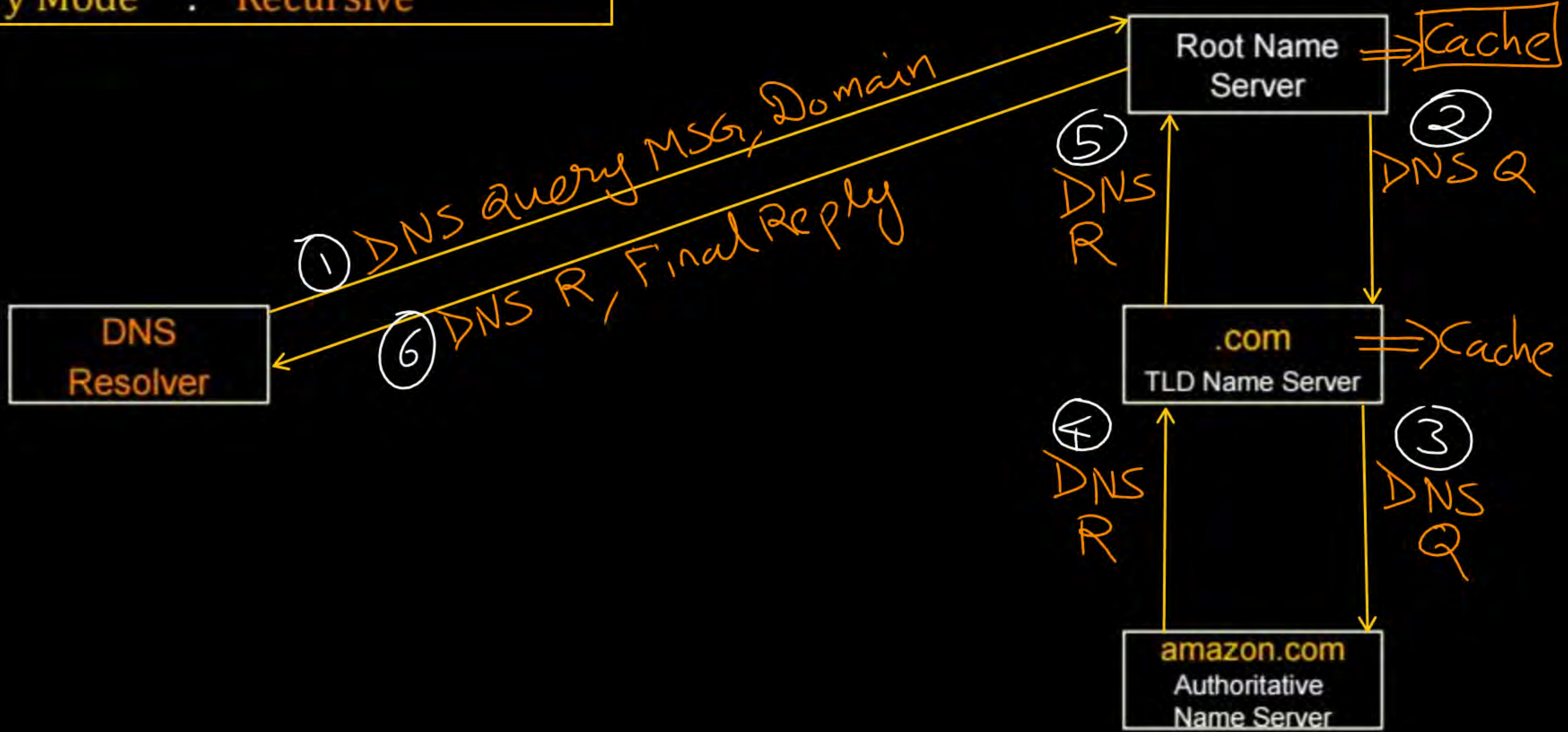
- Put burden of name resolution on Contacted name server [Root N.S.]
- Heavy load at upper levels of hierarchy

Example :-

Domain Name : "www.amazon.com"

Domain Name : "www.amazon.com"
 Query Mode : Recursive

DNS Hierarchy

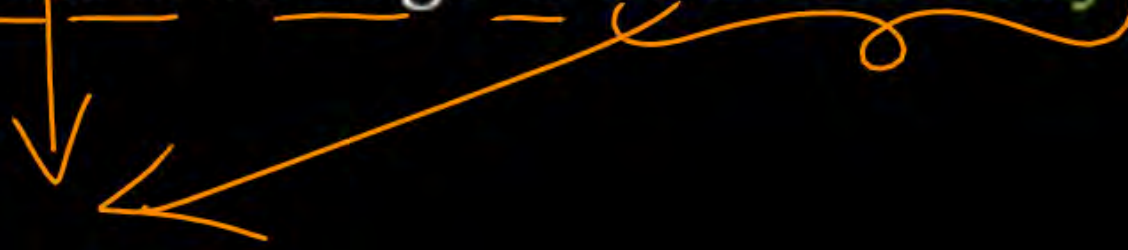




Topic : Local DNS Name Servers



- Each ISP has local DNS name server
- Local DNS server does not belongs to DNS hierarchy

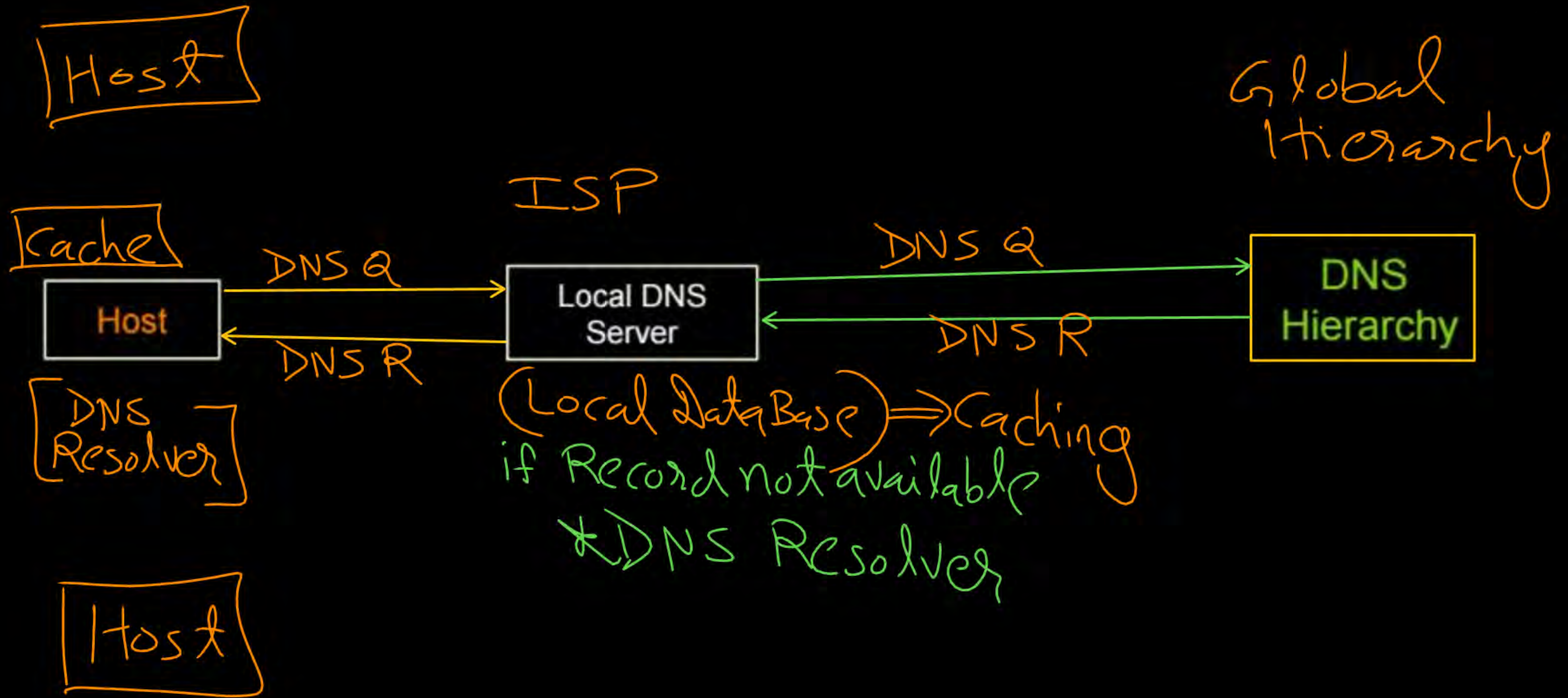




Topic : Local DNS Name Servers



- When host make DNS query, it is sent to its local DNS server
- Local DNS server returns reply
- if Local DNS server does not have valid name-address pair
then it forward request into DNS hierarchy for resolution





Topic : DNS Caching



- Once (any) name server learns mapping, it cache mapping
- Immediately returns a cached mapping in response to a query

=> Caching improves response time

=> Cached entries time-out (disappear) after some time (TTL)



Topic : DNS Records

- Distributed database storing resource records (RR)
- RR format : (Name, Value, Type, TTL)

Type	Name	Value
<u>A</u>	<u>Host Name</u>	<u>IP Address</u>
<u>NS</u>	Domain Name	<u>Authoritative Name Server IP Address</u>
CNAME	<u>Alias Name</u>	<u>Canonical Name</u>
MX	Name	<u>Associated SMTP Mail Server Name</u>



Topic : DNS Message Format

→ DNS query and reply message have same format

Identification	Flags	12 bytes
Number of questions	Number of answer RRs	
Number of authority RRs	Number of additional RRs	
Questions (variable number of questions)		Name, type fields for a query
Answers (variable number of resource records)		RRs in response to query
Authority (variable number of resource records)		Records for authoritative servers
Additional information (variable number of resource records)		Additional "helpful" info that may be used



Topic : DNS



→ DNS uses UDP as transport protocol for query and reply message when message size is less than equal to 512 bytes

→ if DNS messages size is greater than 512 bytes then DNS uses TCP as transport protocol



Topic : DNS



→ if DNS client doesn't get response from DNS server for its query
then DNS uses TCP as transport protocol for query and reply message

→ DNS uses TCP for 'zone transfer'

↓
DNS server data migration



2 mins Summary



Topic

DNS ✓

Topic

~~HTTP~~



THANK - YOU

